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# SOME ERGODIC PROPERTIES OF A COMPLEX CONTINUED FRACTION ALGORITHM

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**ABSTRACT**

Some ergodic properties of a continued fraction algorithm for complex numbers are given.

Recently R. Kaneiwa, J. Tamura and the author [2] proved, by making use of a certain kind of continued fraction algrithm for complex numbers, a theorem of Perron on complex Diophantine approximations [4]: *For any complex number $\theta$ not belonging to the imaginary quadratic field $Q(\sqrt{-3})$ there exist infinitely many integers $p$, $q$ in $Q(\sqrt{-3})$ such that*

$$\left|\theta-\frac{p}{q}\right|<\frac{1}{\sqrt[4]{13}|q|^2}.$$

*If $\theta=\frac{1}{2}(\zeta+\sqrt{\zeta^2+4})$, where $\zeta=\frac{1}{2}(1+\sqrt{-3})$, the constant $\sqrt[4]{13}$ can not be improved.*

In this paper we shall investigate some ergodic properties of the complex continued fractions defined in [2].

## 1. Definition of the algorithm

Every complex number $z$ can be uniquely written in the form $z=u\zeta+v\bar{\zeta}$, where $u$ and $v$ are real, and $\bar{w}$ is the complex conjugate of a complex number $w$. We put

$$[z]=[u]\zeta+[v]\bar{\zeta},$$

where, in the right-hand side, $x$ is the largest rational integer not exceeding a real number $x$. Note that if $z$ is real then $[z]$ becomes the ordinary Gauss' symbol. Thus we define a continued fraction algorithm as follows;

$$\left.\begin{aligned} r_n&=r_n(z)=r_{n-1}^{-1}-[r_{n-1}^{-1}]\ (n\geq 1), \quad r_0=z-[z]\\ a_n&=a_n(z)=[r_{n-1}^{-1}]\ (n\geq 1), \quad a_0=[z]\end{aligned}\right\} \tag{1}$$

These procedures terminate, i.e. $r_n=0$ for some $n\geq 0$, if and only if $z$ belongs to $Q(\sqrt{-3})$. Hence every complex number $z$ can be expanded in the form

$$z=a_0+\cfrac{1}{a_1}+\cdots+\cfrac{1}{a_n+r_n} \qquad (n\geq 0), \tag{2}$$

provided $r_k\neq 0$ for all $k<n$. We put

$$X=\{u\zeta+v\bar{\zeta};\ 0\leq u, v<1\},$$

and define a trnsformation $T$ of $X$ onto itself by

$$Tz=\frac{1}{z}-\left[\frac{1}{z}\right] \quad (z\in X), \tag{3}$$

which is an extension of the well-known '*continued fraction transformation*'

$$Tx=\frac{1}{x}-\left[\frac{1}{x}\right] \quad (x\in[0,1)) \tag{4}$$

(cf. [1]). Thus the remainder $r_n=r_n(z)$ in the algorithm (1) is the $n$th power of the transformation (3) (i.e. $r_n=T^n z$, for all $z\in X$.)

Now we exhibit some basic properties of the algorithm (1). Let $Z_\zeta$ be the ring of all integers in $Q(\sqrt{-3})$ and let $N_\zeta$ be the subset of $Z_\zeta$ defined by

$$N_\zeta=\{a\zeta+b\bar{\zeta};\ a, b \text{ non-negative integers with } a+b\geq 1\}.$$

We put

$$D=\{u\zeta+v\bar{\zeta};\ u, v\geq 0\},$$

and set

$$Y=D\backslash\{z;\ z^{-1}\in X\}.$$

Thus by the definitions we have

$$\begin{aligned}&\{a_0(z);\ z\in C\}=Z_\zeta,\\ &\{a_n(z);\ z\in C\}=N_\zeta=(D\cap Z_\zeta)\backslash\{0\} \quad (n\geq 1),\end{aligned} \tag{5}$$

where $C$ is the set of all complex numbers; and

$$\max_{z\in Y}|z|=\frac{2}{\sqrt{3}}, \tag{6}$$

$$\min_{z\in D\setminus X}|z|=\frac{\sqrt{3}}{2}, \tag{7}$$

Let $a_1, a_2, \cdots, a_n, \cdots$ be any sequence of complex numbers in $D\setminus\{0\}$ not necessarily integral. Every finite continued fraction

$$\frac{1}{|a_1|}+\frac{1}{|a_2|}+\cdots+\frac{1}{|a_n|}$$

is well-defined, since the fractions $a_n^{-1}, a_{n-1}+a_n^{-1}, \cdots$ are different from zero. If, more precisely, $a_n\in D\setminus X$ for all $n\geq 1$, then $a_n^{-1}\in Y\setminus\{0\}$ and so $a_{n-1}+a_n^{-1}\in D\setminus X$. Repeating this process we get

$$a_1+\frac{1}{|a_2|}+\cdots+\frac{1}{|a_n|}\in D\setminus X \tag{8}$$

and

$$\frac{1}{|a_1|}+\frac{1}{|a_2|}+\cdots+\frac{1}{|a_n|}\in Y\setminus\{0\}. \tag{9}$$

Let $a_0\in Z_\zeta$ and $a_n\in N_\zeta$ $(n\geq 1)$. Every finite continued fraction

$$a_0+\frac{1}{|a_1|}+\cdots+\frac{1}{|a_n|}$$

has a canonical representation $p_n/q_n$ $(p_n, q_n\in Z_\zeta)$, called $n$th approximant, in the form of an ordinary fraction. Especially if the sequence $a_0, a_1, \cdots$ is given by the algorithm (1) we call the fraction $p_n/q_n$ the $n$th approximant of $z$. Thus from the general theory of finite continued fractions we have the following formulae: (For the proofs see [5].)

$$p_n=a_np_{n-1}+p_{n-2},\ q_n=a_nq_{n-1}+q_{n-2} \qquad (n\geq 1), \tag{10}$$

$$\frac{1}{|a_n|}+\frac{1}{|a_{n-1}|}+\cdots+\frac{1}{|a_1|}=\frac{q_{n-1}}{q_n} \qquad (n\geq 1), \tag{11}$$

$$p_nq_{n-1}-p_{n-1}q_n=(-1)^{n-1} \qquad (n\geq 0), \tag{12}$$

where $p_{-1}=1$, $q_{-1}=0$, $p_0=a_0$, $q_0=1$. Further if $p_n/q_n$ is the $n$th approximant of $z$, then

$$z-\frac{p_n}{q_n}=(-1)^n\left(a_{n+1}+T^{n+1}z+\frac{q_{n-1}}{q_n}\right)^{-1}q_n^{-2}. \tag{13}$$

**Lemma 1.** (R. Kaneiwa, I. Shiokawa, and J. Tamura [2]) *Let $a_0\in Z_\zeta$ and $a_n\in N_\zeta$ $(n\geq 1)$. Then we have*

$$|q_n|\to\infty \quad as \quad n\to\infty.$$

*where $q_n$ is the denominator of the nth approximant.*

For completeness we prove this lemma.

*Proof.* Suppose, on the contrary, that $q_n \nrightarrow \infty$ as $n \to \infty$. So we can choose an infinite subsequence $\{q_{n_j}\}_{j=1}^{\infty}$ such that $|q_{n_j}| < M$ for all $n \geq 1$, where $M$ is a constant independent of $j$. But from (6) and (9) we have

$$\left|\frac{p_n}{q_n}\right| < \left|a_0 + \frac{2}{\sqrt{3}}\right|$$

and so

$$|p_{n_j}| < \left(|a_0| + \frac{2}{\sqrt{3}}\right)M,$$

where the right-hand side is also independent of $j$. It follows from these inequalities that $p_{n_j}/q_{n_j} = p_{n_k}/q_{n_k}$ for some $j$ and $k$ with $j<k$, since the ring of all integers in $Q(\sqrt{-3})$ is discrete. Hence we have

$$\frac{1}{|a_{n_j}+1|} + \frac{1}{|a_{n_j}+2|} + \cdots + \frac{1}{|a_{n_k}|} = 0,$$

which contradicts (9).

**LEMMA 2.** (ibid.) *Let $z$ be any complex number not belonging to $Q(\sqrt{-3})$ and let $p_n/q_n$ be its nth approximent. Then we have*

$$z = \lim_{n\to\infty} \frac{p_n}{q_n}.$$

*Proof.* By (13) as well as (7), (8), (11) we have

$$\left|z - \frac{p_n}{q_n}\right| < \frac{2}{\sqrt{3}} |q_n|^{-2},$$

which tend to zero as $n \to \infty$.

**LEMMA 3.** (ibid) *With the same notations as in Lemma 1, the nth approximant $p_n/q_n$ converges to some complex number which belongs to $a_0 + Y$.*

*Proof.* Similar to that of Lemma 2.

By means of Lemma 2, every complex number $z$ can be expressed as an infinite regular continued fraction whose partial denominators $a_n(z)$ are integers in $Q(\sqrt{-3})$;

$$z = a_0(z) + \frac{1}{|a_1(z)|} + \frac{1}{|a_2(z)|} + \cdots.$$

This continued fraction expansion is a natural extension of the ordinary real one, since both algorithms coincide when $z$ is real. (For further properties of the algorithm see [2] and [3].)

## 2. Admissible sequences and fundamental cells

We put

$$A^{(n)}=\{a_1(z)\cdots a_n(z);\ z\in X\} \qquad (1\leq n\leq\infty)$$

Sequences belonging to $A^{(n)}$ $(1\leq n\leq\infty)$ will be called admissible. (Note that Lemma 3 suggests the existence of non-admissible sequences.) For any $a_1\cdots a_n\in A^{(n)}$ we define

$$X_{a_1\cdots a_n}=\{z\in X;\ a_k(z)=a_k,\ 1\leq k\leq n\},$$

which will be called a fundamental cell of rank $n$. Thus we have

$$X=\bigcup_{a_1\cdots a_n\in A^{(n)}} X_{a_1\cdots a_n}$$

where $X_{a_1\cdots a_n}\cap X_{b_1\cdots b_n}=\phi$ if $a_k\neq b_k$ for some $k$ with $1\leq k\leq n$; i.e. the set of all fundamental cells of rank $n$ forms a partition of $X$. Besides, for any fixed infinite admissible sequence $a_1a_2\cdots$ we find

$$X\supset X_{a_1}\supset\cdots\supset X_{a_1\cdots a_{n-1}}\supset X_{a_1\cdots a_n}$$

and (by Lemma 2)

$$\operatorname{diam}(X_{a_1\cdots a_n})\to 0 \qquad \text{as } n\to\infty.$$

Hence every Lebesgue measurable subset of $X$ may be approximated with any accuracy by finite unions of mutually disjoint fundamental cells.

For any given $a_1\cdots a_n\in A^{(n)}$ we define a function of $z$ by

$$\psi_{a_1\cdots a_n}(z)=\left|\frac{1}{a_1}\right|+\cdots+\left|\frac{1}{a_{n-1}}\right|+\left|\frac{1}{a_n+z}\right|$$

or equivalently

$$=\frac{p_n+p_{n-1}z}{q_n+q_{n-1}z} \qquad (z\in X). \tag{14}$$

Because of the formula (12) the linear transformation $\psi_{a_1\cdots a_n}$ has the inverse

$$(\psi_{a_1\cdots a_n})^{-1}(z)=\frac{p_n-q_nz}{-p_{n-1}+q_{n-1}z} \qquad (z\in\psi_{a_1\cdots a_n}(X)). \tag{15}$$

But the equality (2) can be rewitten in the form

$$z=\psi_{a_1\cdots a_n}(T^nz) \qquad (z\in X).$$

Hence for each $a_1\cdots a_n\in A^{(n)}$ the $n$th power of $T$ restricted on the cell $X_{a_1\cdots a_n}$ is identical with the inverse of $\psi_{a_1\cdots a_n}$; i.e.

$$T^nz=(\psi_{a_1\cdots a_n})^{-1}(z) \qquad (z\in X_{a_1\cdots a_n}). \tag{16}$$

Especially we have

$$X_{a_1\cdots a_n}=\psi_{a_1\cdots a_n}(T^nX_{a_1\cdots a_n}). \tag{17}$$

Now we need some notations: Put

$$U_1=\left\{z\in X;\ \left|z+\frac{\sqrt{-3}}{3}\right|>\frac{\sqrt{3}}{3}\right\},$$

$$U_2=\{z\in X;\ \mathrm{Im}(z)>0\},$$

$$U_3=\{z\in X;\ \bar{z}\in U_1,\ \mathrm{Im}(z)>0\},$$

and define

$$U_{-j}=\{\bar{z};\ z\in U_j\}\qquad (j=1,2,3).$$

Further we set $U_0=X$ for notational convenience. Considering the reciprocals $U_j^{-1}=\{z;\ z^{-1}\in U_j\}$, we obtain (see Fig. 1)

$$X=\psi_\zeta(U_1)\cup\psi_{\bar{\zeta}}(U_{-1})\cup\left(\bigcup_{\substack{a\in N_\zeta\\ a\neq\zeta,\bar{\zeta}}}\psi_a(X)\right), \tag{18.0}$$

$$U_1=\psi_\zeta(U_{-3})\cup\psi_{\bar{\zeta}}(U_{-1})\cup\left(\bigcup_{k=1}^{\infty}\psi_{\zeta+k}(U_{-2})\right)\cup\left(\bigcup_{\substack{a\in N_\zeta,\ a\neq\bar{\zeta}\\ \mathrm{Im}(a)\leqq 0}}\psi_a(X)\right), \tag{18.1}$$

$$U_2=\psi_{\bar{\zeta}}(U_{-1})\cup\left(\bigcup_{k=1}^{\infty}\psi_k(U_{-2})\right)\cup\left(\bigcup_{\substack{a\in N_\zeta,\ a\neq\zeta\\ \mathrm{Im}(a)<0}}\psi_a(X)\right), \tag{18.2}$$

and

$$U_3=\psi_{\bar{\zeta}}(U_3)\cup\bigcup_{k=1}^{\infty}(\psi_k(U_{-2})\cup\psi_{\bar{\zeta}+k}(U_2)). \tag{18.3}$$

Taking the complex conjugate of (18.1)-(18.3) we have also the same relations for $U_{-1}$, $U_{-2}$, and $U_{-3}$ to which we assign (18.-1), (18.-2), and (18.-3) resp.

In any case $U_j$ can be written in the form

$$U_j=\bigcup_{a\in M_j}\psi_a(U_k) \tag{19}$$

where $M_j$ is a subset of $N_\zeta$ and $k(-3\leq k\leq 3)$ are chosen uniquely according as $j$ and $a$. In addition, we note that

$$\psi_a(X)\cap\psi_b(X)=\phi \tag{20}$$

whenever $a\neq b(a, b\in N_\zeta)$.

**LEMMA 4.** *Let $n\geq 1$ and let $a_1\cdots a_n\in A^{(n)}$. Then we have*

$$X_{a_1\cdots a_n}=\psi_{a_1\cdots a_n}(U_j), \tag{21}$$

*and so*

$$T^nX_{a_1\cdots a_n}=U_j \tag{22}$$

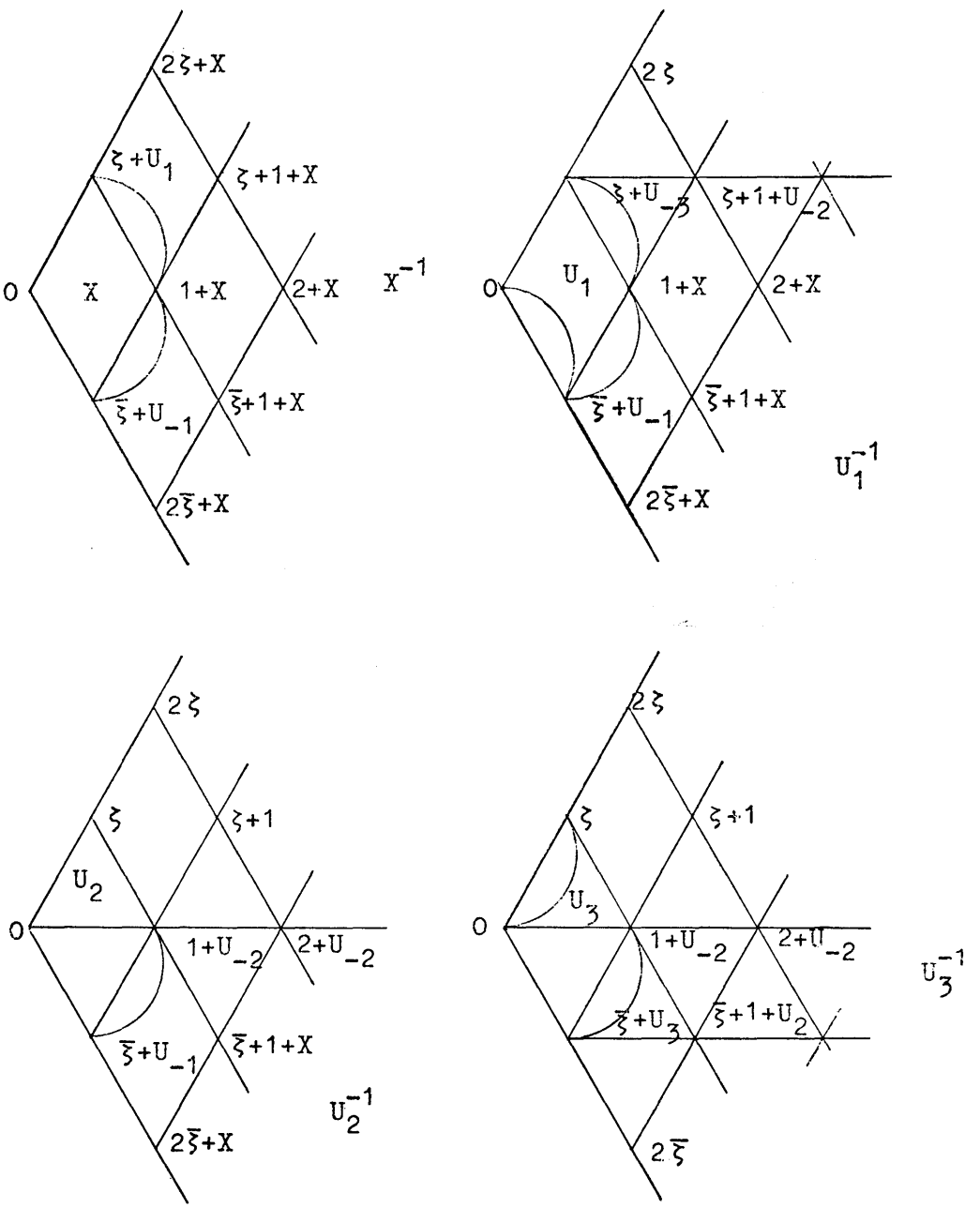

2ζ+X
ζ+U1
ζ+1+X
0
X
1+X
2+X
X^-1
ξ̄+U-1
ξ̄+1+X
2ξ̄+X
2ζ
ζ+U-3
ζ+1+U-2
U1
1+X
2+X
ξ̄+U-1
ξ̄+1+X
U1^-1
2ξ̄+X
2ζ
ζ
ζ+1
U2
0
1+U-2
2+U-2
ξ̄+U-1
ξ̄+1+X
U2^-1
2ξ̄+X
2ζ
ζ
ζ+1
U3
0
1+U-2
2+U-2
U3^-1
ξ̄+U3
ξ̄+1+U2
2ξ̄

Fig. 1

*for some* $j$ $(-3\leq j\leq 3)$.

*Proof.* By induction on $n$. First we prove (21). If $n=1$ (21) follows from (18.0). Suppose that (21) holds for all $a_1\cdots a_n\in A^{(n)}$. Then we have

$$\begin{aligned}X_{a_1\cdots a_{n-1}}&=\{z\in X_{a_1\cdots a_n};\ a_{n-1}(z)=a_1(T^n z)=a_{n+1}\}\\&=\{\psi_{a_1\cdots a_n}(w);\ w\in U_j, a_1(w)=a_{n+1}\}\\&=\psi_{a_1\cdots a_n}(\psi_{a_{n+1}}(U_k)),\ \text{(by (14), (15))}\\&=\psi_{a_1\cdots a_{n+1}}(U_k),\end{aligned}$$

for $a_1\cdots a_{n+1}\in A^{(n+1)}$, where $j$ is defined by $U_j=T^n X_{a_1\cdots a_n}$ and $k$ chosen uniquely in (19). (22) follows from (17) and (21).

Let $E$ be any subset of $X$. Then by Lemma 4 we have

$$\begin{aligned}T^{-n}E&=\{z\in X;\ T^n z\in E\}\\&=\bigcup_{a_1\cdots a_n\in A^{(n)}}\{z\in X_{a_1\cdots a_n};\ T^n z\in E\cap T^n X_{a_1\cdots a_n}\}\\&=\bigcup_{a_1\cdots a_n\in A^{(n)}}\psi_{a_1\cdots a_n}(E\cap T^n X_{a_1\cdots a_n}).\end{aligned}\tag{23}$$

## 3. Estimates of the Lebesgue measure

Let $m$ be the Lebesgue measure on the complex plane and let $\mathfrak{B}$ be the $\sigma$-field of all measurable subsets of $X$. Then we have for any $a_1\cdots a_n\in A^{(n)}$ and any $E\in\mathfrak{B}$

$$m(\psi_{a_1\cdots a_n}(E))=\iint_E|\psi'_{a_1\cdots a_n}(z)|^2dxdy,\qquad z=x+iy.\tag{24}$$

But using (12) we find

$$|\psi'_{a_1\cdots a_n}(z)|^2=|q_n|^{-4}\left|1+\frac{q_{n-1}}{q_n}z\right|^{-4}.\tag{25}$$

Hence

$$3^{-4}<|q_n|^4|\psi'_{a_1\cdots a_n}(z)|^2<3^4\tag{26}$$

and

$$3^{-4}<|q_n|^{-4}|(\psi^{-1}_{a_1\cdots a_n})'(z)|^2<3^4,\tag{27}$$

because (from (6), (7), (9), (11))

$$3^{-1}<\frac{\sqrt{3}}{2}\leq\left|1+\frac{q_{n-1}}{q_n}z\right|\leq 1+\frac{2}{\sqrt{3}}<3.\tag{28}$$

(23) and (25) give the estimates

$$3^{-4}m(E)<|q_n|^4m(\psi_{a_1\cdots a_n}(E))<3^4m(E)\qquad(E\in\mathfrak{B})\tag{29}$$

Eepecially, taking account of the fact that $3^{-2}<m(U_j)<1$ $(-3\leq j\leq 3)$, we have

$$3^{-6}<|q_n|^4 m(X_{a_1\cdots a_n})<3^4 \qquad (a_1\cdots a_n\in A^{(n)}). \tag{30}$$

We write

$$S(n)=\sum_{a_1\cdots a_n\in A^{(n)}}|q_n|^{-4}.$$

Then for any $n\geq 1$ we have

$$3^{-5}<S(n)<3^6. \tag{31}$$

Indeed it follows from (30) that

$$3^4 S(n)>\sum_{A^{(n)}} m(X_{a_1\cdots a_n})=m(X)>3^{-1}$$

and

$$3^{-6}S(n)<m(X)<1.$$

By means of Lemma 4 the set $A^{(n)}$ of all admissible sequences can naturally be divided into seven subsets; we put

$$A_j^{(n)}=\{a_1\cdots a_n\in A(n);\ T^n X_{a_1\cdots a_n}=U_j\} \qquad (-3\leq j\leq 3),$$

then

$$A^{(n)}=\bigcup_{j=-3}^{3} A_j^{(n)}.$$

By (18.j) $(-3\leq j\leq 3)$ we have the following relations for $n>1$;

$$\begin{aligned} A_0^{(n)}=\{a_1\cdots a_n\in A^{(n)};\ & a_1\cdots a_{n-1}\in A_0^{(n-1)},\ a_n\neq\zeta,\bar{\zeta};\\ \text{or } & a_1\cdots a_{n-1}\in A_1^{(n-1)},\ a_n\neq\bar{\zeta},\ \mathrm{Im}\,(a_n)\leq 0;\\ \text{or } & a_1\cdots a_{n-1}\in A_{-1}^{(n-1)},\ a_n\neq\zeta,\ \mathrm{Im}\,(a_n)\geq 0;\\ \text{or } & a_1\cdots a_{n-1}\in A_2^{(n-1)},\ a_n\neq\bar{\zeta},\ \mathrm{Im}\,(a_n)<0;\\ \text{or } & a_1\cdots a_{n-1}\in A_{-2}^{(n-1)},\ a_n\neq\zeta,\ \mathrm{Im}\,(a_n)>0\}, \end{aligned} \tag{32.0}$$

$$A_1^{(n)}=\{a_1\cdots a_n\in A^{(n)};\ a_1\cdots a_{n-1}\in A_0^{(n-1)}\cup A_{-1}^{(n-1)}\cup A_{-2}^{(n-1)},\ a_n=\zeta\}, \tag{32.1}$$

$$\begin{aligned} A_2^{(n)}=\{a_1\cdots a_n\in A^{(n)};\ & a_1\cdots a_{n-1}\in A_{-1}^{(n-1)}\cup A_3^{(n-1)},\ a_n-\bar{\zeta}\in N;\\ \text{or } & a_1\cdots a_{n-1}\in A_{-2}^{(n-1)}\cup A_{-3}^{(n-1)},\ a_n\in N\}, \end{aligned} \tag{32.2}$$

$$A_3^{(n)}=\{a_1\cdots a_n\in A^{(n)};\ a_1\cdots a_{n-1}\in A_{-1}^{(n-1)}\cup A_{33}^{(n-1)},\ a_n=\bar{\zeta}\}, \tag{32.3}$$

and

$$A_{-j}^{(n)}=\{\bar{a}_1\cdots\bar{a}_n;\ a_1\cdots a_n\in A_j^{(n)}\} \qquad (j=1,2,3), \tag{32.-j}$$

where $N$ is the set of all positive integers.

We write

$$S_j(n)=\sum_{a_1\cdots a_n\in A_j^{(n)}}|q_n|^{-4} \qquad (-3\leq j\leq 3).$$

Thus we have

$$S_j(n)=S_{-j}(n) \qquad (-3\leq j\leq 3), \tag{33}$$

and

$$S(n)=\sum_{j=-3}^{3}S_j(n). \tag{34}$$

**LEMMA 5.** *For any* $j$ $(-3\leq j\leq 3)$ *we have*

$$S_j(n)>3^{-12} \qquad (n\geq 3). \tag{35}$$

*Proof.* Let $a_1\cdots a_n\in A^{(n)}$ and let $p_n/q_n$ the $n$ th approximant. Then it follows from (10) and (28) that

$$3^{-1}|a_nq_{n-1}|<|q_n|<3|a_nq_{n-1}|. \tag{36}$$

By (32.0), (33) and (36) we have

$$\begin{aligned}3S_0(n)>&\sum_{a\in N_\zeta\setminus\{\zeta,\bar\zeta\}}|a|^{-4}\sum_{a_1\cdots a_{n-1}\in A_0^{(n-1)}}|q_{n-1}|^{-4}\\&+2\sum_{\substack{a\in N_\zeta\setminus\{\zeta\}\\ \mathrm{Im}(a)\leq 0}}|a|^{-4}\sum_{a_1\cdots a_{n-1}\in A_1^{(n-1)}}|q_{n-1}|^{-4}\\&+2\sum_{\substack{a\in N_\zeta\setminus\{\bar\zeta\}\\ \mathrm{Im}(a)<0}}|a|^{-4}\sum_{a_1\cdots a_{n-1}\in A_2^{(n-1)}}|q_{n-1}|^{-4}\\>&S_0(n-1)+2S_1(n-1)+2|\bar\zeta+1|^{-4}S_2(n-1)\\>&3^{-2}(S_0(n-1)+S_1(n-1)+S_2(n-1)).\end{aligned}$$

Hence

$$S_0(n)>3^{-3}(S_0(n-1)+S_1(n-1)+S_2(n-1)).$$

In the same way we get

$$S_1(n)>3^{-1}(S_0(n-1)+S_1(n-1)+S_2(n-1)),$$

$$S_2(n)>3^{-1}(S_1(n-1)+S_2(n-1)+S_3(n-1)),$$

and

$$S_3(n)>3^{-1}(S_1(n-1)+S_3(n-1))$$

(using (32.1)-(32.3)). These inequalities as well as (31), (33), and (34) imply that

$$S_0(n)>3^{-6}\sum_{j=0}^{3}S_j(n-2)>3^{-7}S(n)>3^{-12}.$$

Similarly we may obtain (35) for any $j(-3\leq j\leq 3)$.

In what follows we shall use (35) only with $j=0$.

## 4. Invariant measure and ergodicity

**Theorem 1.** *Let $E$ be any measurable subset of $X$ such that $T^{-1}E=E$. Then $m(E)=0$ or $m(X)$.*

*Proof.* We assume that $m(E)>0$. By Lemma 4 and (23) we find for any $a_1\cdots a_n\in A^{(n)}$

$$\begin{aligned}E\cap X_{a_1\cdots a_n}&=T^{-n}E\cap\psi_{a_1\cdots a_n}(T^nX_{a_1\cdots a_n})\\&=\psi_{a_1\cdots a_n}(E\cap T^nX_{a_1\cdots a_n}).\end{aligned}$$

From this as well as (29) and (30) we have

$$\begin{aligned}m(E\cap X_{a_1\cdots a_n})&\geq 3^{-4}|q_n|^{-4}m(E\cap T^nX_{a_1\cdots a_n})\\&\geq 3^{-8}m(X_{a_1\cdots a_n})\min\{m(E\cap U_3),m(E\cap U_{-3})\}.\end{aligned}\tag{37}$$

But (18.3) and (23) implies that

$$E\cap U_3=T^{-1}E\cap U_3\supset\psi_{\xi+1}(E\cap U_2)\cup\psi_1(E\cap U_{-2}).$$

Beside for any measurable subset $F$ of $U_2$ we have by (24) with (25)

$$m(\psi_1(F))=\iint_F|1+z|^{-4}dxdy>\iint_F|\bar{\xi}+1+z|^{-4}dxdy=m(\psi_{\xi-1}(F)).$$

Hence

$$\begin{aligned}m(E\cap U_3)&>m(\psi_{\xi+1}(E\cap U_2))+m(\psi_{\xi+1}(E\cap U_{-2}))\\&=m(\psi_{\xi+1}(E))>3^{-4}|\bar{\xi}+1|^{-4}m(E)=3^{-6}m(E).\end{aligned}\tag{38}$$

(using (29)). Similarly we get

$$m(E\cap U_{-3})>3^{-6}m(E).\tag{39}$$

By (37), (38), and (39) the inequality

$$m(E\cap F)\geq 3^{-14}m(E)m(F).\tag{40}$$

hold for all fundamental cell $F$, and so for any measurable set $F$ in $X$. Thus, putting $F=X\setminus E$ in (40), we have

$$m(E)m(X\setminus E)=0,$$

which implies $m(E)=m(X)$.

**Theorem 2.** *There exists an unique, $T$-invariant probability measure $\mu$ equivalent to Lebesgue measure such that the inequalities*

$$3^{-15}\frac{m(E)}{m(X)}\leq\mu(E)\leq 3^{10}\frac{m(E)}{m(X)}\tag{41}$$

*hold for all* $E\in\mathfrak{B}$.

*Proof.* To prove the existence it is enough to show that the inequalities

$$3^{-15}m(E)<m(T^{-n}E)<3^{10}m(E) \qquad (E\in\mathfrak{B}) \tag{42}$$

hold for all $n\geq 0$ (see F. Schweiger [5] §6-§7). By (23), (29), and (31) we have

$$\begin{aligned} m(T^{-n}E) &< \sum_{A^{(n)}} m(\psi_{a_1\cdots a_n}(E)) \\ &\leq 3^4 m(E)S(n)\leq 3^{10}m(E) \end{aligned}$$

To prove the left-hand side inequalities in (42), we suppose first that $E\subset U_3$. Then, by (23), (29), and Lemma 5,

$$m(T^{-n}E)\geq \sum_{A_0^{(n)}} m(\psi_{a_1\cdots a_n}(E))\geq 3^{-16}m(E),$$

as required. In the same way, these inequalities hold for any $E\subset U_2\setminus U_3$. The left-hand side of the inequalities (42) is also true for any subset $E$ of $U_{-3}$ or $U_{-2}\setminus U_{-3}$. As a result (42) holds for any subset $E$ of $X$, since

$$E=(E\cap U_3)\cup(E\cap(U_2\setminus U_3))\cup(E\cap U_{-2})\cup(E\cap(U_{-2}\setminus U_{-3})).$$

By Theorem 1 $T$-invariant probability measure $\mu$ is uniquely given by the limit

$$\mu(E)=\frac{1}{m(X)}\lim_{n\to\infty}\frac{1}{n}\sum_{k=0}^{n-1} m(T^{-k}E) \qquad (E\in\mathfrak{B}) \tag{43}$$

(see F. Schweiger [5]). So (41) follows from (42) and (43).

**THEOREM 3.** *$T$ is ergodic with respect to $\mu$; i.e. for any $f\in L^1(X)$ we have*

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=0}^{n-1} f(T^k z)=\int_X f(z)d\mu, \quad a.e.$$

*Proof.* Follows from Theorem 1, 2 and Birkhoff's individual ergodic theorem. As an application of Theorem 3, we have

$$\lim_{n\to\infty}(a_1(z)\cdots a_n(z))^{1/n}=e^{\alpha}, \quad a.e.$$

where

$$\alpha=\int_X \log a_1(z)d\mu.$$

(Note that $f(z)=\log a_1(z)\in L^1(X)$, where $-\frac{\pi}{3}\leq \arg f(z)\leq\frac{\pi}{3}$, since the series $\sum_{a\in N_\zeta} a^{-4}\log a$ is convergent.)

## 5. Exactness

A measure-preserving transformation $T$ on a normalized measure space $(X, \mathfrak{B}, \mu)$

is said to be *exact* if

$$\bigcap_{n=0}^{\infty} T^{-n}\mathfrak{B}=\{\phi, X\},$$

or equivalently, if for every set $E$ of positive measure with the measurable images $TE, T^2E, \cdots$ the relation

$$\lim_{n\to\infty} \mu(T^nE)=1 \tag{44}$$

holds (see V.A. Rohlin [6].)

**Theorem 4.** *The transformation $T$ is exact.*
The proof requires the following

**Lemma 6.** *Let $\varepsilon>0$ and let $E$ be any measurable set such that*

$$\mu(U_j\backslash E)<\varepsilon$$

*for some $j$ $(-3\leq j\leq 3)$. Then*

$$\mu(TE)>1-3^{31}\varepsilon.$$

*Proof of Lemma* 6. It is clearly enough to consider only the case $j=\pm 3$. Further we may assume $j=3$, since the following arguments are available for the conjugate case $j=-3$. Note first that

$$\phi_1(U_{-2})\cup\phi_{\xi+1}(U_2)\subset U_3, \tag{45}$$

(by (18.3)). By (41) and (27) with $n=1$ and $a_n=1$, we get

$$\begin{aligned}\mu(T(\phi_1(U_{-2})\backslash E))&\leq 3^{10}m(X)^{-1}m(T(\phi_1(U_{-2})\backslash E))\\&\leq 3^{14}m(X)^{-1}m(\phi_1(U_{-2})\backslash E)\leq 3^{29}\mu(\phi_1(U_{-2})\backslash E).\end{aligned} \tag{46}$$

while, using (27) with $n=1$ and $a_1=\xi+1$,

$$\mu(T(\phi_{\xi+1}(U_2)\backslash E))\leq 3^{31}\mu(\phi_{\xi+1}(U_2)\backslash E)). \tag{47}$$

Combining (45), (46), and (47) we find

$$\begin{aligned}&\mu(T((\phi_1(U_{-2})\cup\phi_{\xi+1}(U_2))\backslash E))\\&\quad\leq 3^{31}\mu((\phi_1(U_{-2})\cup\phi_{\xi+1}(U_2))\backslash E)\\&\quad\leq 3^{31}\mu(U_3\backslash E)\leq 3^{31}\varepsilon.\end{aligned} \tag{48}$$

Therefore, by (45), (48), and (18.3), we obtain

$$\begin{aligned}\mu(TE)&\geq\mu(T((\phi_1(U_{-2})\cup\phi_{\xi+1}(U_2))\backslash E))\\&\geq\mu(T(\phi_1(U_{-2})\cup\phi_{\xi+1}(U_2)))-\mu(T((\phi_1(U_{-2})\cup\phi_{\xi+1}(U_2))\backslash E))\\&>1-3^{31}\varepsilon.\end{aligned}$$

*Proof of Theorem 3.* We prove (44). Let $E\in\mathfrak{B}$ given arbitrary. (Note that, by the definition of $T$, $E\in\mathfrak{B}$ if and only if $TE\in\mathfrak{B}$.) Let $\varepsilon>0$. Then there exists a

fundamental interval $F=X_{a_1\cdots a_n}$ such that

$$m(F\setminus E)<3^{-50}\varepsilon m(F). \tag{49}$$

Otherwise, the inequality

$$m(F\setminus E)\geq 3^{-50}\varepsilon m(F)$$

holds for all fundamental interval $F$, and so it holds also for arbitrary measurable set $F$. Putting $F=E$ we have $m(F)=0$; a contradiction.

Using Lemma 4, (41), (16), (27), (30), and (49), we get

$$\begin{aligned}\mu(T^nF\setminus T^nE)&\leq\mu(T^n(F\setminus E))\\&\leq 3^{11}m(T^n(F\setminus E))\leq 3^{15}|q_n|^4m(F\setminus E)\\&\leq 3^{19}m(F)^{-1}m(F\setminus E)<3^{-31}\varepsilon.\end{aligned} \tag{50}$$

Noticing that $T^nF=U_j$ for some $j$ by Lemma 4, we have from (50) and Lemma 6

$$\mu(T^{n+1}E)>1-\varepsilon.$$

Since $\mu(E), \mu(TE), \mu(T^2E), \cdots$ is non-decreasing, the relation (44) is proved.

As a general property of exact transformations (see V.A. Rohlin [4]) we have

**Corollary.** *The transformation $T$ is mixing of all degrees. In particular $T$ is strongly mixing; i.e. for any $E, F\in\mathfrak{B}$ we have*

$$\lim_{n\to\infty}\mu(T^{-n}E\cap F)=\mu(E)\mu(F).$$

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